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CIRCUITS AND METHODS FOR SENSING CURRENT IN RESONANT TANKS OF SWITCHING POWER SUPPLIES

Abstract

A circuit is disclosed. The circuit includes a first semiconductor switch connected to a second semiconductor switch at a switch node, an input voltage coupled to the first semiconductor switch, a ground coupled to the second semiconductor switch, a resonant circuit coupled to the first and second semiconductor switches, the resonant circuit having an inductor in series with a resonant capacitor coupled to the input voltage, and a sense resistor, where a voltage across the sense resistor is proportional to a current in the resonant circuit. In one aspect, the circuit further includes a sense capacitor coupled between the resonant capacitor and the sense resistor. In another aspect, inductor is coupled in parallel with a resonant capacitor.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Chinese provisional patent application no. 202410238639.7, for “METHOD FOR SENSING RESONANT TANK CURRENT OF SWITCHING POWER SUPPLY” filed on Mar. 1, 2024, which is hereby incorporated by reference in entirety for all purposes.

FIELD

[0002] The described embodiments relate generally to power converters, and more particularly, the present embodiments relate to systems and methods for sensing current in resonant tanks of switching power supplies.

BACKGROUND

[0003] With the development of power electronics, the industry has put forward higher requirements for the power density of a switching power supply. Soft switching technology can enable a power device to achieve zero-voltage switching-on (ZVS) and zero-current switching-off (ZCS), effectively reducing the switching loss of the converter. Current soft switching topologies may include a resonant tank, that is, a network composed of a resonant capacitor $C_{sub.r}$ and a resonant inductor $L_{sub.r}$. Through the resonance of $C_{sub.r}$ and $L_{sub.r}$, a power device can be enabled to achieve ZVS and ZCS. For soft switching topologies, accurate and efficient sensing of the resonant tank current is useful.

SUMMARY

[0004] In some embodiments, a circuit is disclosed. The circuit includes a first semiconductor switch connected to a second semiconductor switch at a switch node; an input voltage coupled to the first semiconductor switch; a ground coupled to the second semiconductor switch; a resonant circuit coupled to the first and second semiconductor switches, the resonant circuit having an inductor in series with a resonant capacitor coupled to the input voltage; and a sense resistor, where a voltage across the sense resistor is proportional to a current in the resonant circuit.

[0005] In some embodiments, the circuit further includes a sense capacitor coupled between the resonant capacitor and the sense resistor.

[0006] In some embodiments, the resonant circuit includes a transformer having a primary side and a secondary side.

[0007] In some embodiments, the resonant capacitor is coupled to a drain terminal of the first semiconductor switch.

[0008] In some embodiments, the input voltage is coupled to the drain terminal of the first semiconductor switch.

[0009] In some embodiments, the sense resistor is coupled between the resonant capacitor and the ground.

[0010] In some embodiments, the sense capacitor is a first sense capacitor and where the circuit further includes a second sense capacitor coupled to the sense resistor.

[0011] In some embodiments, the circuit further includes a control circuit coupled to the sense resistor and arranged to sense the voltage across the resistor.

[0012] In some embodiments, a circuit is disclosed. The circuit includes a first semiconductor switch connected to a second semiconductor switch at a switch node; an input terminal coupled to the first semiconductor switch; a ground coupled to the second semiconductor switch; a resonant circuit coupled to the first and second semiconductor switches, the resonant circuit having an inductor in parallel with a resonant capacitor coupled to the input terminal; and a sense resistor

coupled to the second semiconductor switch, where a voltage across the sense resistor is proportional to a current in the resonant circuit.

[0013] In some embodiments, a method of operating a circuit is disclosed. The method includes: providing a first semiconductor switch; providing a second semiconductor switch coupled to the first semiconductor switch at a switch node; providing an input terminal coupled to the first semiconductor switch; providing a ground coupled to the second semiconductor switch; providing a resonant circuit coupled to the first and second semiconductor switches, the resonant circuit having an inductor in series with a resonant capacitor coupled to the input terminal; and sensing a voltage across a sense resistor that is proportional to a current in the resonant circuit, where the sense resistor is coupled to the second semiconductor switch.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is a schematic of a half-bridge LLC converter with components for sensing a resonant tank current according to certain aspects of the present disclosure;

[0015] FIG. 2 is a set of plots that demonstrate a resonant tank current of a half-bridge LLC converter according to certain aspects of the present disclosure;

[0016] FIG. 3 is a schematic of a half-bridge flyback converter with components for sensing a resonant tank current according to certain aspects of the present disclosure;

[0017] FIG. 4 is a set of plots that demonstrate a resonant tank current of a asymmetric half-bridge flyback converter according to certain aspects of the present disclosure;

[0018] FIG. 5 is a schematic of an active clamp flyback converter with components for sensing a resonant tank current according to certain aspects of the present disclosure;

[0019] FIG. 6A is a schematic of a system for sensing a resonant tank current according to certain aspects of the present disclosure; and

[0020] FIG. 6B is a schematic of an alternative system for sensing a resonant tank current according to certain aspects of the present disclosure.

DETAILED DESCRIPTION

[0021] Circuits, devices and related techniques disclosed herein relate generally to power converters. More specifically, circuits, devices and related techniques disclosed herein relate to systems and methods for sensing current in resonant tanks of switching power supplies.

Embodiments of the disclosure enable use of lossless resonant tank current sensing on the high side in power converter topologies such as, but not limited to, asymmetric half-bridge (AHB), inductor-inductor-capacitor (LLC) half-bridge LLC converters, and active-clamp-flyback (ACF) converters. Circuits and techniques disclosed herein can detect a resonant tank current in switching power supplies relatively accurately and with relatively high efficiency by use of a passive network that can include a sensing capacitor and a sensing resistor.

[0022] In current approaches, current sensing solutions can use a sensing resistor or a sensing capacitor. The sensing resistor method can cause ohmic losses and can reduce efficiency of the converter. The sensing capacitor method can be lossless, however it may work on the low-side only since the sensing capacitor is connected to the power ground. Thus, the capacitor sensing method can only be used in topologies with the transformer of the half-bridge converter being disposed on the low side. Embodiments of the disclosure can enable sensing of a resonant tank current on the high side as well as on the low side. In some embodiments, a controller circuit can use the sensed current for power converter loop control to provide for stable and reliable operation of the convert, and to prevent overcurrent in the power switches. Various inventive embodiments are described herein, including methods, processes, systems, devices, and the like.

[0023] Several illustrative embodiments will now be described with respect to the accompanying

drawings, which form a part hereof. The ensuing description provides embodiment(s) only and is not intended to limit the scope, applicability, or configuration of the disclosure. Rather, the ensuing description of the embodiment(s) will provide those skilled in the art with an enabling description for implementing one or more embodiments. It is understood that various changes may be made in the function and arrangement of elements without departing from the spirit and scope of this disclosure. In the following description, for the purposes of explanation, specific details are set forth in order to provide a thorough understanding of certain inventive embodiments. However, it will be apparent that various embodiments may be practiced without these specific details. The figures and description are not intended to be restrictive. The word “example” or “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment or design described herein as “exemplary” or “example” is not necessarily to be construed as preferred or advantageous over other embodiments or designs.

[0024] In current approaches, asymmetric half-bridge (AHB), inductor-inductor-capacitor (LLC) half-bridge LLC converters, and active-clamp-flyback (ACF) topologies may be used in power converter circuits. In current approaches, during start-up and load-shedding process of in a half-bridge LLC converter, the resonant tank current can appear to have an excessive amplitude, and the converter can have a risk of entering a capacitive region. Embodiments of the disclosure enable sensing of the resonant tank current to ensure safe and stable operation of the converter. The resonant tank can be referred to as a resonant cavity

[0025] In current approaches, current sensing solutions can be either a resistor sensing or capacitor sensing. The resistor sensing method can cause ohmic losses and can reduce efficiency of the converter. The capacitor sensing method can be a lossless current sensing method, however, can be used on the low side only since one end of the resonant capacitor $C_{sub.r}$ is connected to a power ground. In current approaches, relatively complex control circuit for capacitive sensing is used in power converters, thus increasing costs. Embodiment of the disclosure enable relatively simple control scheme thereby saving costs.

[0026] In current approaches, when an output voltage of the AHB flyback converter drops, due to a mismatch between a voltage of the resonant capacitor $V_{sub.cr}$ and the output voltage $V_{sub.o}$, a large current stress can occur in the resonant tank, which may further cause over-current damage to a synchronous control switch. Therefore, the synchronous control switch of the AHB flyback converter may require cycle-by-cycle overcurrent protection. The resonant tank of the AHB flyback converter can be disposed on the high side or the low side. In addition, the AHB flyback converter can work in a peak current control mode, using a controller circuit to sense a peak current of a main control switch. Embodiments of the disclosure enable use of switching devices with a lossless current sensing function, such as Navitas' GaNsense™ technology, that are disposed on the low side or on the high side.

[0027] In current approaches, during dynamic testing of a ACF power converter, the resonant tank of the ACF converter may experience a large current stress. Embodiments of the disclosure can enable safe and reliable test and operation of ACF converters.

[0028] In some embodiments, the resonant capacitor $C_{sub.r}$ and the resonant inductor $L_{sub.r}$ can form a resonant tank of a switching power supply. The resonant inductor may be an independent inductor or a leakage inductance of a transformer. In the half-bridge LLC converter, the resonant inductor $L_{sub.r}$ can generally be an independent inductor. In the AHB flyback converter and the ACF converter, the resonant inductor can generally be a leakage inductance of a transformer. In various embodiments, one end of the resonant capacitor $C_{sub.r}$ can be directly connected to the input voltage, and the other end thereof can be connected to the sensing capacitor.

[0029] In some embodiments, the sensing capacitor and the sensing resistor can be connected in series to form a passive RC network. By sensing a difference between input voltage and voltage of a resonant capacitor, a current value of the resonant tank can be accurately obtained. Embodiment of the disclosure enable use of a voltage of the sensing resistor to represent current information of a

resonant tank, and a calculation formula is as follows:

$$[00001] \quad V_{\text{sense}} = C_{\text{sense}} \frac{d(V_{\text{in}} \pm V_{\text{cr}})}{dt} R_{\text{sense}} \quad (1)$$

where C.sub.sense is a capacitance of the sensing capacitor, R.sub.sense is the resistance of the sensing resistor, V.sub.sense is the voltage of the sensing resistor, V.sub.in is the input voltage, and V.sub.cr is the voltage of the resonant capacitor. In formula (1), the half-bridge LLC converter and the asymmetric half-bridge flyback converter are marked with a '-' sign. Formula (1) can be rewritten as formula (2):

$$[00002] \quad V_{\text{sense}} = C_{\text{sense}} \frac{d(V_{\text{in}} - V_{\text{cr}})}{dt} R_{\text{sense}} \quad (2)$$

In formula (1), the active clamp flyback converter is marked with a '+' sign. Formula (1) can be rewritten as formula (3):

$$[00003] \quad V_{\text{sense}} = C_{\text{sense}} \frac{d(V_{\text{in}} + V_{\text{cr}})}{dt} R_{\text{sense}} \quad (3)$$

In the present disclosure, the sensing resistor R.sub.sense can be connected in parallel with a capacitor or diode to further regulate amplitude and phase of the voltage V.sub.sense of the sensing resistor.

[0030] In order to make the technical advantages of the present disclosure clear, circuits and methods for sensing a resonant tank current of switching power supplies are described in detail below with reference to the accompanying drawings. In particular, the embodiments described herein are merely used to explain the present disclosure and are not intended to limit the present disclosure.

[0031] FIG. 1 is a schematic of a half-bridge LLC converter **100** with components for sensing a resonant tank current, according to certain aspects of the present disclosure. The half-bridge LLC converter **100** can include a transformer **102**, a first switch **106** (labeled Q.sub.1 in FIG. 1), a second switch **104** (labeled Q.sub.2 in FIG. 1), a resonant capacitor **108** (labeled C.sub.r in FIG. 1), a sense capacitor **110** (labeled C.sub.sense in FIG. 1), a sense resistor **112** (labeled R.sub.sense in FIG. 1), and an LLC controller **114**. The half-bridge LLC converter **100** can also include a number of nodes, such as node **101**, node **103**, node **105**, node **107**, and node **109**. Additionally, the LLC converter **100** can include a voltage input V.sub.in and a ground location PGND. Since the second switch **104** is closer to the voltage input V.sub.in than the first switch **106**, the second switch **104** can be referred to as a high side switch and the first switch **106** can be referred to as a low side switch. In the half-bridge LLC converter **100**, the first switch **106** can act as a main switch and the second switch **104** can act as a synchronous switch. In some examples, the first switch **106** can operate without a bootstrap power supply.

[0032] The transformer **102** can include a primary side and a secondary side. The primary side of the transformer **102** can include a resonant inductor (labeled L.sub.r in FIG. 1). The primary side of the transformer **102** and the resonant capacitor **108** can form the resonant tank. The resonant tank can be referred to as a resonant cavity. Current that flows through the resonant tank can be referred to as a tank current. The resonant tank can be connected to leads (e.g., lead **101** and lead **107**) adjacent to the second switch **104** or high side switch. Thus, the half-bridge LLC converter **100** can have an arrangement with a high side resonant tank.

[0033] One end of the resonant capacitor **108** can be connected to node **103**. Another end of the resonant capacitor **108** can be connected to the voltage input V.sub.in via the lead **101**. The connection to voltage input V.sub.in can act as a 'static point' for the resonant capacitor **108**. The tank current can be sensed easier when the resonant capacitor **108** has the "static point". A first end of the sense capacitor **110** can be connected to node **103** and a second end of the sense capacitor **110** can be connected to node **105**. A first end of the sense resistor **112** can be connected to node **105** and second end of the sense resistor **112** can be connected to node **109**. A voltage across the sense resistor **112** can be called a sense voltage (labeled V.sub.sense in FIG. 1). The LLC controller **114** can be connected to node **105**.

[0034] The sense capacitor **110** and the sense resistor **112** can be connected in series to form a passive RC network. By sensing a difference between the input voltage $V_{\text{sub.in}}$ and voltage of the resonant capacitor **108**, a value for the tank current of the resonant tank can be accurately obtained. The voltage $V_{\text{sub.sense}}$ across the sense resistor **112** can represent tank current information of the resonant tank, and a calculation formula for $V_{\text{sub.sense}}$ is as follows:

$$[00004] \quad V_{\text{sense}} = C_{\text{sense}} \frac{d(V_{\text{in}} - V_{\text{cr}})}{dt} R_{\text{sense}} \quad (4)$$

where $C_{\text{sub.sense}}$ is a capacitance of the sense capacitor **110**, $R_{\text{sub.sense}}$ is the resistance of the sense resistor **112**, $V_{\text{sub.sense}}$ is the voltage of the sense resistor **112**, $V_{\text{sub.in}}$ is the input voltage, and $V_{\text{sub.cr}}$ is the voltage of the resonant capacitor **108**.

[0035] In LLC converter **100**, a resonant tank is disposed on a high side of the half-bridge LLC converter, $Q_{\text{sub.1}}$ can act as a main control switch, and $Q_{\text{sub.2}}$ can act as a synchronous control switch. Embodiment of the disclosure enable direct connection of one end of a resonant capacitor $C_{\text{sub.r}}$ to an input voltage $V_{\text{sub.in}}$, and another end thereof to a sensing capacitor **110** labeled $C_{\text{sub.sense}}$. Through differential sampling of the sensing capacitor and a sensing resistor, embodiments of the disclosure can detect current of the resonant tank with relative accuracy and efficiency.

[0036] FIG. 2 shows graphs of sensed current in the tank of the half-bridge LLC converter **100**, according to some embodiments. For FIG. 2, the sensing capacitor **110** $C_{\text{sub.sense}}$ has a value of, for example, 1 nF, and sensing resistor **112** $R_{\text{sub.sense}}$ has a value of, for example, 15Ω. It can be seen from FIG. 2 that the circuits and methods disclosed herein can accurately sense the current of the resonant tank and convert the resonant tank current into a voltage $V_{\text{sub.sense}}$ of the sensing resistor. During a startup process of the half-bridge LLC converter, a peak current value of the resonant tank can be very large. At this time, a large voltage $V_{\text{sub.sense}}$ can also appear across the sensing resistor. After a control circuit detects the resonant tank current by sensing the voltage $V_{\text{sub.sense}}$, the controller can respond relatively quickly to turn off the main switch, thereby improving the reliability of the half-bridge LLC converter. Embodiments of the disclosure can enable sensing of the resonant tank current cycle by cycle, thereby allowing the sensed current to be used as an input parameter of a control loop of the half-bridge LLC converter.

[0037] FIG. 3 is a schematic of an AHB flyback converter **300** with components for sensing a resonant tank current according to certain aspects of the present disclosure. The AHB flyback converter **300** can include a transformer **302**, a first switch **306** (labeled $Q_{\text{sub.1}}$ in FIG. 3), a second switch **304** (labeled $Q_{\text{sub.2}}$ in FIG. 3), a resonant capacitor **308** (labeled $C_{\text{sub.r}}$ in FIG. 3), a sense capacitor **310** (labeled $C_{\text{sub.sense}}$ in FIG. 3), a sense resistor **312** (labeled $R_{\text{sub.sense}}$ in FIG. 3), and an AHB controller **314**. The AHB flyback converter **300** can also include a number of nodes, such as node **301**, node **303**, node **305**, and node **307**. Additionally, the AHB flyback converter **300** can include a voltage input $V_{\text{sub.in}}$ and a ground location PGND. Since the second switch **304** is closer to the voltage input $V_{\text{sub.in}}$ than the first switch **306**, the second switch **304** can be referred to as a high side switch and the first switch **306** can be referred to as a low side switch. In the AHB flyback converter **300**, the first switch **306** can act as a main switch and the second switch **304** can act as a synchronous switch. In some examples, the first switch **306** can operate without a bootstrap power supply.

[0038] The transformer **302** can include a primary side and a secondary side. The primary side of the transformer **302** can include a resonant inductor (labeled L in FIG. 3). The primary side of the transformer **302** and the resonant capacitor **308** can form the resonant tank. Current that flows through the resonant tank can be referred to as a tank current. The resonant tank can be connected to leads (e.g., lead **301** and lead **307**) adjacent to the second switch **304** or high side switch. Thus, the AHB flyback converter **300** can have an arrangement with a high side resonant tank.

[0039] One end of the resonant capacitor **308** can be connected to node **303**. Another end of the resonant capacitor **308** can be connected to the voltage input $V_{\text{sub.in}}$ via the lead **301**. The

connection to voltage input V.sub.in can act as a ‘static point’ for the resonant capacitor **308**. The tank current can be sensed easier when the resonant capacitor **308** has the “static point”. A first end of the sense capacitor **310** can be connected to node **303** and a second end of the sense capacitor **310** can be connected to node **305**. A first end of the sense resistor **312** can be connected to node **305** and a second end of the sense resistor **312** can be connected to ground. A voltage across the sense resistor **312** can be called a sense voltage (labeled V.sub.sense in FIG. 3). The AHB controller **314** can be connected to node **305**.

[0040] The sense capacitor **310** and the sense resistor **312** can be connected in series to form a passive RC network. By sensing a difference between the input voltage V.sub.in and voltage of the resonant capacitor **308**, a value for the tank current of the resonant tank can be accurately obtained. The voltage V.sub.sense across the sense resistor **312** can represent tank current information of the resonant tank, and a calculation formula for V.sub.sense is as follows:

$$[00005] \quad V_{\text{sense}} = C_{\text{sense}} \frac{d(V_{\text{in}} - V_{\text{cr}})}{dt} R_{\text{sense}} \quad (5)$$

where C.sub.sense is a capacitance of the sense capacitor **310**, R.sub.sense is the resistance of the sense resistor **312**, V.sub.sense is the voltage of the sense resistor **312**, V.sub.in is the input voltage, and V.sub.cr is the voltage of the resonant capacitor **308**. The AHB controller **314** can continuously monitor the tank current. In some examples, the AHB controller **314** can compare the monitored value of the tank current to a threshold value and can automatically turn off a switch, such as the second switch **304**, if the monitored value exceeds the threshold value. The AHB controller **314** can protect switches from overload currents.

[0041] In some embodiments, the asymmetric half-bridge flyback converter **300** can operate in a peak current control mode, that uses sensing of the current in the main control switch. When the first switch acts as the main control switch, lossless current sensing of the power device can be used to improve efficiency of the converter. Embodiments of the disclosure enable connection of one end of a resonant capacitor C.sub.r to an input voltage V.sub.in and enable detecting current information of the resonant tank accurately and efficiently through a sensing capacitor and a sensing resistor.

[0042] FIG. 4 shows graphs of sensed current in the tank of the asymmetric half-bridge flyback converter **300**, according to some embodiments. For FIG. 4, the sensing capacitor **310** C.sub.sense has a value of, for example, 20 pF, and sensing resistor **312** R.sub.sense has a value of, for example, 2 KΩ. It can be seen from FIG. 4 that the circuits and methods disclosed herein can accurately sense the current of the resonant tank. When an output voltage of the asymmetric half-bridge flyback converter drops, a pulse current exceeding 16 A may appear in the resonant tank, which may cause damage to the second switch. By using the sensing method disclosed herein, after detecting the current information of the resonant tank, a controller circuit can compare the voltage V.sub.sense with an overcurrent protection setting V.sub.ocp. When the controller circuit finds that V.sub.sense > V.sub.ocp, the controller circuit can switch off the second switch relatively rapidly to prevent damage to the power switch.

[0043] FIG. 5 is a schematic of an ACF converter **500** with components for sensing a resonant tank current according to certain aspects of the present disclosure. The ACF converter **500** can include a transformer **502**, a first switch **506** (labeled Q.sub.1 in FIG. 5), a second switch **504** (labeled Q.sub.2 in FIG. 5), a resonant capacitor **508** (labeled C.sub.r in FIG. 5), a sense capacitor **510** (labeled C.sub.sense in FIG. 5), a sense resistor **512** (labeled R.sub.sense in FIG. 5), and an ACF controller **514**. The ACF flyback converter **500** can also include a number of nodes, such as node **501**, node **503**, node **505**, node **507**, and node **509**. Additionally, the ACF converter **500** can include a voltage input V.sub.in and a ground location PGND. Since the second switch **504** is closer to the voltage input V.sub.in than the first switch **506**, the second switch **504** can be referred to as a high side switch and the first switch **506** can be referred to as a low side switch. In the ACF converter **500**, the first switch **506** can act as a main switch and the second switch **504** can act as a

synchronous switch. In some examples, the first switch **506** can operate without a bootstrap power supply.

[0044] The transformer **502** can include a primary side and a secondary side. The primary side of the transformer **502** can include a resonant inductor (labeled L.sub.r in FIG. 5). The primary side of the transformer **502** and the resonant capacitor **508** can form the resonant tank. The resonant tank can be referred to as a resonant cavity. Current that flows through the resonant tank can be referred to as a tank current. The resonant tank can be connected to leads (e.g., lead **503** and lead **507**) adjacent to the second switch **504** or high side switch. Thus, the ACF converter **500** can have an arrangement with a high side resonant tank.

[0045] One end of the resonant capacitor **508** can be connected to node **503**. Another end of the resonant capacitor **508** can be connected to the voltage input V.sub.in via the lead **501**. The connection to voltage input V.sub.in can act as a ‘static point’ for the resonant capacitor **508**. The tank current can be sensed easier when the resonant capacitor **508** has the “static point”. A first end of the sense capacitor **510** can be connected to node **503** and a second end of the sense capacitor **510** can be connected to node **505**. A first end of the sense resistor **512** can be connected to node **505** and a second end of the sense resistor **512** can be connected to node **509**. A voltage across the sense resistor **512** can be called a sense voltage (labeled V.sub.sense in FIG. 5). The ACF controller **514** can be connected to node **505**.

[0046] The sense capacitor **510** and the sense resistor **512** can be connected in series to form a passive RC network. By sensing a sum of the input voltage V.sub.in and voltage of the resonant capacitor **508**, a value for the tank current of the resonant tank can be accurately obtained. The voltage V.sub.sense across the sense resistor **512** can represent tank current information of the resonant tank, and a calculation formula for V.sub.sense is as follows:

$$[00006] \quad V_{\text{sense}} = C_{\text{sense}} \frac{d(V_{\text{in}} + V_{\text{cr}})}{dt} R_{\text{sense}} \quad (6)$$

where C.sub.sense is a capacitance of the sense capacitor **510**, R.sub.sense is the resistance of the sense resistor **512**, V.sub.sense is the voltage of the sense resistor **512**, V.sub.in is the input voltage, and V.sub.cr is the voltage of the resonant capacitor **508**. The ACF controller **514** can continuously monitor the tank current. In some examples, the ACF controller **514** can compare the monitored value of the tank current to a threshold value and can automatically turn off a switch, such as the second switch **504**, if the monitored value exceeds the threshold value. The ACF controller **514** can protect switches from overload currents. Embodiments of the disclosure enable connection of one end of a resonant capacitor **508** C.sub.r to the input voltage V.sub.in, and can enable detecting current of the resonant tank accurately and efficiently through a sensing capacitor **510** and a sensing resistor **512**.

[0047] FIG. 6A is a schematic of a system **600** for sensing a resonant tank current according to certain aspects of the present disclosure. The system **600** can be similar to LLC converter **100** from FIG. 1, AHB flyback converter **300** from FIG. 3, or ACF converter **500** from FIG. 5. The system **600** can have an arrangement that includes a high side resonant tank. Additionally, the system **600** can include sensing components that can measure a tank current (e.g. resonant tank current). The sensing components can include a first sense capacitor **610** (labeled C.sub.sense1 in FIG. 6A), a sense resistor **612** (labeled R.sub.sense in FIG. 6A), a second sense capacitor **616** (labeled C.sub.sense2 in FIG. 6A), and a controller **614**. The controller **614** can be LLC controller **114**, AHB controller **314**, or ACF controller **514**. The sensing components can be similar to the passive RC networks shown in FIGS. 1, 3 and 5, but the sensing components can include the additional second sense capacitor **616**. The second sense capacitor **616** can be connected in parallel with the sense resistor **612**.

[0048] The sensing components can allow the controller **614** to measure (using e.g., equations (4), (5), or (6) as described above) and monitor the resonant tank current via a sensed voltage across the sense resistor **612**. In some examples, the controller **614** can use the measured resonant tank

current to protect components, such as a high side switch, by comparing the measured current to a threshold current value. The controller **614** can switch components off if the measured resonant tank current exceeds the threshold current value. The second sense capacitor **616** can enable an amplitude or phase adjustment of the sense voltage. The amplitude or phase adjustment can cause the sense voltage to more closely align with the resonant tank current, simplifying and improving a capability of the controller **614** to control the system **600**.

[0049] FIG. **6B** is a schematic of an alternative system **650** for sensing a resonant tank current according to certain aspects of the present disclosure. The system **650** can be similar to LLC converter **100** from FIG. **1**, AHB flyback converter **300** from FIG. **3**, or ACF converter **500** from FIG. **5**. The system **650** can have an arrangement that includes a high side resonant tank. Additionally, the system **650** can include sensing components that can measure a tank current (e.g. resonant tank current). The sensing components can include a sense capacitor **660** (labeled C.sub.sense1 in FIG. **6B**), a sense resistor **662** (labeled R.sub.sense in FIG. **6B**), a diode **668** (labeled D.sub.sense in FIG. **6B**), and a controller **664**. The controller **664** can be LLC controller **114**, AHB controller **314**, or ACF controller **514**. The sensing components can be similar to passive RC networks shown in FIGS. **1**, **3** and **5**, but the sensing components can include the diode **668**. The diode **668** can be connected in parallel with the sense resistor **662**.

[0050] The sensing components can allow the controller **664** to measure (using e.g., equation (4), (5), or (6) as described above) and monitor the resonant tank current via a sensed voltage across the sense resistor **662**. In some examples, the controller **664** can use the measured resonant tank current to protect components, such as a high side switch, by comparing the measured current to a threshold current value. The controller **664** can switch components off if the measured resonant tank current exceeds the threshold current value. The diode **668** can enable an amplitude or phase adjustment of the sense voltage. The amplitude or phase adjustment can cause the sense voltage to more closely align with the resonant tank current, simplifying and improving a capability of the controller **664** to control the system **650**. In some examples, the adjustment can eliminate negative values of the sense voltage and protect components of the controller **664** that may be damaged by negative voltages.

[0051] In the foregoing specification, embodiments of the disclosure have been described with reference to numerous specific details that can vary from implementation to implementation. The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense. The sole and exclusive indicator of the scope of the disclosure, and what is intended by the applicants to be the scope of the disclosure, is the literal and equivalent scope of the set of claims that issue from this application, in the specific form in which such claims issue, including any subsequent correction. The specific details of particular embodiments can be combined in any suitable manner without departing from the spirit and scope of embodiments of the disclosure.

[0052] Additionally, spatially relative terms, such as “bottom” or “top” and the like can be used to describe an element and/or feature's relationship to another element(s) and/or feature(s) as, for example, illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the switch in use and/or operation in addition to the orientation depicted in the figures. For example, if the switch in the figures is turned over, elements described as a “bottom” surface can then be oriented “above” other elements or features. The switch can be otherwise oriented (e.g., rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

[0053] Terms “and,” “or,” and “an/or,” as used herein, may include a variety of meanings that also is expected to depend at least in part upon the context in which such terms are used. Typically, “or” if used to associate a list, such as A, B, or C, is intended to mean A, B, and C, here used in the inclusive sense, as well as A, B, or C, here used in the exclusive sense. In addition, the term “one or more” as used herein may be used to describe any feature, structure, or characteristic in the singular or may be used to describe some combination of features, structures, or characteristics. However, it

should be noted that this is merely an illustrative example and claimed subject matter is not limited to this example. Furthermore, the term “at least one of” if used to associate a list, such as A, B, or C, can be interpreted to mean any combination of A, B, and/or C, such as A, B, C, AB, AC, BC, AA, AAB, ABC, AABBBCCC, etc.

[0054] Reference throughout this specification to “one example,” “an example,” “certain examples,” or “exemplary implementation” means that a particular feature, structure, or characteristic described in connection with the feature and/or example may be included in at least one feature and/or example of claimed subject matter. Thus, the appearances of the phrase “in one example,” “an example,” “in certain examples,” “in certain implementations,” or other like phrases in various places throughout this specification are not necessarily all referring to the same feature, example, and/or limitation. Furthermore, the particular features, structures, or characteristics may be combined in one or more examples and/or features.

[0055] In the preceding detailed description, numerous specific details have been set forth to provide a thorough understanding of claimed subject matter. However, it will be understood by those skilled in the art that claimed subject matter may be practiced without these specific details. In other instances, methods and apparatuses that would be known by one of ordinary skill have not been described in detail so as not to obscure claimed subject matter. Therefore, it is intended that claimed subject matter not be limited to the particular examples disclosed, but that such claimed subject matter may also include all aspects falling within the scope of appended claims, and equivalents thereof.

Claims

1. A circuit comprising: a first semiconductor switch connected to a second semiconductor switch at a switch node; an input voltage coupled to the first semiconductor switch; a ground coupled to the second semiconductor switch; a resonant circuit coupled to the first and second semiconductor switches, the resonant circuit comprising an inductor in series with a resonant capacitor coupled to the input voltage; and a sense resistor, wherein a voltage across the sense resistor is proportional to a current in the resonant circuit.
2. The circuit of claim 1, further comprising a sense capacitor coupled between the resonant capacitor and the sense resistor.
3. The circuit of claim 1, wherein the resonant circuit comprises a transformer having a primary side and a secondary side.
4. The circuit of claim 1, wherein the resonant capacitor is coupled to a drain terminal of the first semiconductor switch.
5. The circuit of claim 4, wherein the input voltage is coupled to the drain terminal of the first semiconductor switch.
6. The circuit of claim 1, wherein the sense resistor is coupled between the resonant capacitor and the ground.
7. The circuit of claim 2, wherein the sense capacitor is a first sense capacitor and wherein the circuit further comprises a second sense capacitor coupled to the sense resistor.
8. The circuit of claim 1, further comprising a control circuit coupled to the sense resistor and arranged to sense the voltage across the resistor.
9. A circuit comprising: a first semiconductor switch connected to a second semiconductor switch at a switch node; an input terminal coupled to the first semiconductor switch; a ground coupled to the second semiconductor switch; a resonant circuit coupled to the first and second semiconductor switches, the resonant circuit comprising an inductor in parallel with a resonant capacitor coupled to the input terminal; and a sense resistor coupled to the second semiconductor switch, wherein a voltage across the sense resistor is proportional to a current in the resonant circuit.
10. The circuit of claim 9, further comprising a sense capacitor coupled between the resonant

capacitor and the sense resistor.

11. The circuit of claim 9, wherein the resonant circuit comprises a transformer having a primary side and a secondary side.

12. The circuit of claim 9, wherein the resonant capacitor is coupled to a drain terminal of the first semiconductor switch.

13. The circuit of claim 12, wherein the input terminal is coupled to the drain terminal of the first semiconductor switch.

14. The circuit of claim 9, wherein the sense resistor is coupled between the resonant capacitor and the ground.

15. The circuit of claim 10, wherein the sense capacitor is a first sense capacitor and wherein the circuit further comprises a second sense capacitor coupled to the sense resistor.

16. A method of operating a circuit, the method comprising: providing a first semiconductor switch; providing a second semiconductor switch coupled to the first semiconductor switch at a switch node; providing an input terminal coupled to the first semiconductor switch; providing a ground coupled to the second semiconductor switch; providing a resonant circuit coupled to the first and second semiconductor switches, the resonant circuit comprising an inductor in series with a resonant capacitor coupled to the input terminal; and sensing a voltage across a sense resistor that is proportional to a current in the resonant circuit, wherein the sense resistor is coupled to the second semiconductor switch.

17. The method of claim 16, further comprising providing a sense capacitor that is coupled between the resonant capacitor and the sense resistor.

18. The method of claim 16, wherein the resonant circuit comprises a transformer having a primary side and a secondary side.

19. The method of claim 16, wherein the resonant capacitor is coupled to a drain terminal of the first semiconductor switch.

20. The method of claim 19, wherein the input terminal is coupled to the drain terminal of the first semiconductor switch.
